Unified Agent Benchmark (HIL-Bench): Canonical Testing Methods for Unified Intelligence and the Harness Intelligence Level

Chengshuai Yang Ting Xue

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NextGen PlatformAI C Corp, USA
Correspondence: spiritai@platformai.org

Abstract

The companion paper, *Unified Intelligence Theory and the Harness Intelligence Level*, defines a cumulative intelligence hierarchy built from Cognitive Intelligence C, Individual Intelligence I, Organizational Intelligence O, Delegation Intelligence DI, and operational Self-Awareness SA, with directly measured coordinates $[C, I, O, T, H, SA]$. It preserves the gated Unified Intelligence hierarchy $U0$-$U\Omega$ and clarifies Individual Intelligence as the longitudinal evolution of one persistent decision locus: transient operation ($I0$), persistent continuity ($I1$), adaptive learning under a fixed learning mechanism ($I2$), governed self-improvement ($I3$), recursive improvement of improvement ($I4$), persistent autonomous discovery ($I5$), and open-ended expansion of future cognitive reach ($I\Omega$). The theory further proposes an invariance principle: after a standard is ratified, models should move through the scale; the meaning of the scale should not move with the models.

This paper defines the corresponding **measurement standard**. HIL-Bench separates an invariant **canonical test law** from the concrete task instances used to obtain evidence. For each coordinate level, the standard specifies the experimental intervention, allowed resources, human-intervention budget, external verifier, lower-level retention requirements, reliability criterion, and admissible evidence. A fixed public reference set makes every level inspectable and reproducible. A fixed private anchor set supplies a stable historical certification reference. New private witness forms may be added for contamination resistance only when they implement the same canonical test law; they extend evidence without redefining the level. Some upper-level claims use frozen human-review protocols when objective automatic verification is insufficient, but the rubric, blinding, admissibility criteria, and acceptance rule are part of the standard rather than discretionary judgment.

HIL-Bench evaluates two subjects. In **agent mode**, a frozen model-harness pair is measured directly and receives a gated Unified level, the continuous pair score $HLIS$, the coordinate profile $[C, I, O, T/H, SA]$, delegation frontiers, false-completion rate, uncertainty, and a declared resource envelope. In **model mode**, one frozen base LLM is evaluated through a standardized reference harness ladder, yielding a Harness Scaling Curve, HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, verification responsiveness ρ_V, and an optional composite HIL-Score. The curve and gated level remain primary.

The main contribution is therefore not another moving benchmark corpus. It is a candidate **stable experimental constitution for intelligence measurement**: level semantics are defined in Paper 1; canonical experiments are defined here; datasets are witnesses to those experiments. Benchmark saturation motivates higher levels or additional witness forms, not movement of an existing level boundary.

Keywords: Harness Intelligence Level; HIL-Bench; Unified Intelligence; agent benchmark; LLM benchmark; persistence; adaptive learning; self-improvement; recursive improvement; discovery; public-private evaluation; canonical testing

1. Introduction

Most AI benchmarks are deliberately narrow. Multiple-choice science benchmarks measure answer accuracy; coding benchmarks execute programs against tests; repository benchmarks verify patches; interactive benchmarks measure behavior in an environment. These instruments are useful precisely because their scoring rules are concrete. They become less informative when a single result is interpreted as a complete intelligence level.

The companion theory takes the opposite direction: it defines a structured intelligence hierarchy and asks what evidence is required to support each claim. That creates a measurement problem. If the test changes every time frontier models improve, then an intelligence level becomes a moving percentile rather than a stable scientific concept. If the exact task corpus alone defines a level, then contamination eventually destroys the meaning of the level. If the evaluator is allowed to improvise the harness, human assistance, or acceptance rule, then results from different laboratories are not commensurable.

HIL-Bench addresses these problems by distinguishing three objects:

$$\boxed{\text{Level semantics} \neq \text{Canonical test law} \neq \text{Test form.}}$$

Paper 1 defines the level semantics. This paper defines the canonical test law. A public or private dataset is a test form that instantiates the law.

This separation permits long-term stability. A future model may make an $I1$ restart test trivial, but $I1$ still means persistent individual continuity across a genuine execution discontinuity. A new private restart form may be introduced because an old secret leaked, but the intervention and acceptance rule remain the same. Saturation therefore changes where the research frontier lies, not what the earlier level meant.

The benchmark has four design goals:

1. **Invariant meaning.** A named level has the same operational meaning after ratification, independent of contemporary frontier capability.
2. **External falsifiability.** Every claimed level admits an experiment whose decisive evidence is outside the candidate's write set.
3. **Contamination resistance without moving the ruler.** New hidden forms may be generated under the same law without changing the level definition.
4. **Practical repeatability.** Lower levels should be largely automatic, fast, inexpensive, and reproducible, while upper levels may use longer horizons or expert review where the construct requires it.

This manuscript is a **pre-ratification candidate HIL Measurement Standard 1.0**. Its purpose is to make the proposed ruler explicit enough to test before freezing it.

2. The HIL Measurement Constitution

2.1 Canonical test law

For coordinate d and level n, define the canonical testing law

$$\Pi_{d,n} = (E_{d,n}, X_{d,n}, A_{d,n}, V_{d,n}, R_{d,n}, H_{d,n}, p_{d,n}, K_{d,n}),$$

where:

- $E_{d,n}$ is the admissible environment or task-family class;
- $X_{d,n}$ is the required experimental intervention or lifecycle perturbation;
- $A_{d,n}$ is the allowed action/tool interface and authority envelope;
- $V_{d,n}$ is the external verification rule;
- $R_{d,n}$ is the declared resource envelope;
- $H_{d,n}$ is the maximum permitted human cognitive intervention when relevant;
- $p_{d,n}$ is the reliability requirement;
- $K_{d,n}$ is the lower-level retention set.

A concrete form F supplies particular tasks, seeds, hidden states, perturbations, and keys consistent with $\Pi_{d,n}$. The form is evidence for the level; it is not the definition of the level.

A system A satisfies a coordinate-level claim on form F only if the canonical verifier passes and all required retention evidence passes:

$$A \vDash_F d_n \quad \Longleftrightarrow \quad V_{d,n}(A, E_{d,n}, X_{d,n}, F) = 1 \ \wedge \ K_{d,n} = 1.$$

A certification claim additionally requires the preregistered reliability and uncertainty rule to be satisfied.

2.2 What becomes frozen after ratification

Object	HIL-1.0 treatment
Meanings of C, I, O, T, H, SA	Frozen
Meanings of $U0$-$U\Omega$	Frozen
Canonical test law $\Pi_{d,n}$ for each level	Frozen
Human-intervention semantics $H0$-$H5$	Frozen
Outcome semantics, including false completion	Frozen
Public canonical reference forms	Frozen and permanently public
Private canonical anchor forms	Frozen and permanently archived private
Scoring semantics attached to HIL-1.0	Frozen
New private witness forms	May be added if construct-equivalent
New diagnostic tests	May be added without changing level meaning
Improved implementation of the same test law	Allowed if equivalence is demonstrated
Fundamental change to a level or test law	Requires a new major standard

The rule is:

$$\boxed{\text{New tests may extend evidence; they may not silently redefine an old intelligence level.}}$$

2.3 Three kinds of change

After ratification, changes are classified before results are affected.

Implementation repair. A parser, sandbox, verifier, or logging bug is corrected while the semantic rule remains unchanged. The defect and affected runs are disclosed; the level definition does not change.

Witness renewal. A private form is suspected to be contaminated, leaked, or overfit. A new hidden witness form is generated under the same canonical law. Historical results retain their form identifier; new certifications use the new form.

Semantic revision. Evidence shows that the level predicate or canonical test law itself is conceptually wrong. This is not patched inside HIL-1.0. It motivates HIL-2.0 or another named major standard.

3. Experimental Subjects and the Unit of Measurement

3.1 Agent mode

The experimental unit is a frozen pair

$$A = (m, h),$$

where m is the frozen base model and h is the frozen harness, including memory, prompts, tools, retry logic, role structure, acceptance mechanism, and other runtime machinery. The complete resource and authority envelope is declared.

The primary report for a singleton is

$$\boxed{UI^* \mid HLIS \mid [C, I, T/H, SA]},$$

and for an organization

$$\boxed{UO^* \mid HLIS \mid [C, I, O, T/H, SA]}.$$

A diagnostic memory profile may be reported separately, but memory is not an additional Unified Intelligence coordinate and does not promote I by itself.

3.2 Base-model mode

A raw base LLM does not possess a unique persistent harness. To measure how effectively a model can exploit progressively richer architecture, the same frozen model is evaluated through a standardized harness ladder:

$$HSC_m(k) = HLIS(m, HG_k).$$

The full Harness Scaling Curve is the primary model result. Summary quantities include HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, ρ_V, and an optional HIL-Score.

3.3 Independence of certification

The candidate system may design strategies, artifacts, procedures, or even candidate harness changes when the level permits it. It may not author, modify, select, or reveal the hidden certification tasks, hidden verifier, acceptance thresholds, or independent promotion decision used to certify that change.

The general self-certification rule is:

$$\boxed{\text{the system under test must not control the criterion that declares it successful.}}$$

4. Canonical Testing Methods for Cognitive Intelligence

Cognitive Intelligence isolates problem solving from persistence, organization, and longitudinal self-change. HIL-Bench therefore evaluates C in a minimal standardized harness with controlled tools and context. The exact task content may vary across witness forms; the cognitive demand and verifier semantics do not.

Level	Canonical test method
$C0$ Reactive	Present explicit atomic transformations with exact or deterministic verification. No hidden planning requirement.
$C1$ Contextual	Present routine well-specified problems requiring correct interpretation of ordinary context and a short procedure.
$C2$ Compositional	Present structured tasks requiring multiple facts, constraints, or dependent reasoning steps; use exact constraint or execution verifiers.
$C3$ Strategic	Present unfamiliar tasks requiring strategy construction, causal/abstract reasoning, or transfer to a new surface form; include held-out structural variants.
$C4$ Expert	Present difficult, ambiguous, multi-constraint expert problems with adversarially relevant distractors and independent expert or deterministic verification.
$C5$ Discovery	Present a bounded problem in which the solution method or explanatory model is not given; require a new method/model that predicts sealed cases.
$C6$ Frontier-Generalizing	Present interconnected cross-domain cognitive problems whose solution requires integrating several domains under one controlled episode.

Level	Canonical test method
$C\Omega$ Open-Ended	Longitudinal evidence that the system creates new representations or cognitive methods that expand the classes of bounded problems it can subsequently solve.

To avoid mistaking specialization for general cognition, HIL-1.0 uses a fixed family structure for the common cognitive panel: general, science, mathematics, abstract reasoning, code, multimodal/spatial reasoning, and long-context reasoning. Each family has public reference forms and private witness forms. A score from one family cannot substitute for another family merely because both are difficult.

5. Canonical Testing Methods for Individual Intelligence

Individual Intelligence is longitudinal. It measures one persistent decision locus, not a static model's one-shot reasoning score. The central methodological requirement is causal evidence across lifecycle discontinuities.

5.1 I0: transient operation

$I0$ requires no persistence claim. The system must execute within a bounded current episode under the declared harness. Passing C or delegation tasks does not imply $I1$.

5.2 I1: persistent individual continuity

Capability being tested. Task-relevant individual continuity survives a genuine discontinuity of active execution.

Canonical method I1-RST.

1. Establish hidden task-relevant state, identity facts, commitments, or partial work during episode A.
2. Permit the system to persist information only through declared durable mechanisms.
3. Terminate the active executor process and invalidate ephemeral context.
4. Start a fresh executor process under the same declared identity and resource contract.
5. Present episode B without restating the hidden prior state.
6. Require correct resumption and provenance-linked recovery.
7. Run an ablated control in which episode A did not occur or its persistent state is unavailable.

Certification requires the experienced arm to succeed and the ablated arm not to explain the result. Replaying the entire previous conversation or preserving the same live process does not satisfy the discontinuity requirement.

5.3 I2: adaptive learning under a fixed learning mechanism

Capability being tested. Verified experience causes a persistent, appropriate change in later behavior while the learning mechanism remains fixed.

Canonical method I2-XFER.

$$\text{baseline} \rightarrow \text{verified experience} \rightarrow \text{persistence} \rightarrow \text{restart} \rightarrow \text{held-out transfer}.$$

The task family contains at least two arms: an experienced arm and an ablated no-experience control. The held-out evaluation changes surface form so that copying the exact training episode is insufficient. The learning algorithm, prompt-update rule, consolidation rule, and allowed memory machinery are frozen before the run.

A useful causal quantity is

$$\Delta_{I2} = P(\text{held-out success} \mid \text{verified experience}) - P(\text{held-out success} \mid \text{ablated}).$$

$I2$ is not awarded merely because the system remembers that a failure happened. The experience must alter later behavior in an appropriate direction.

5.3A Canonical Memory Capability Benchmarks (M-Bench)

Memory Capability M is independently measured and then used as a prerequisite gate for Individual Intelligence. It is not a sixth conceptual intelligence family and high memory alone does not promote I. For a tested pair A, let $M(A)$ be its highest independently certified memory level. Each M_k claim requires the level-specific verifier, preregistered reliability/uncertainty rule, and lower-memory retention.

$$\boxed{A \models M_k \iff V_k^M(A) = 1 \ \wedge \ K_{<k}^M(A) = 1}.$$

M0-EPH: ephemeral working state

Capability. Correct use of task-relevant state inside one bounded active execution episode; no durability claim is made.

Canonical method M0-EPH.

1. Present a short sequence containing hidden current-episode variables, commitments, or intermediate results.
2. Query those variables later in the same live episode with distractors and reordered surface form.
3. Verify exact/structured use of the active state under the declared context/tool budget.
4. A restart may be used as a negative diagnostic: loss after restart is compatible with $M0$ and must not be mislabeled $M1$.

M1-RST: persistent memory across restart

Capability. Declared durable state survives a genuine execution discontinuity.

Canonical method M1-RST.

1. In episode A, require explicit persistence of hidden task state using only declared durable mechanisms.
2. Snapshot the memory manifest, terminate the active executor, and invalidate ephemeral context.
3. Start a fresh executor under the same identity/harness contract without replaying episode A.
4. Query the stored state and its provenance; verify exact or semantically equivalent recovery.
5. Run an ablated arm with the relevant durable state removed. Success must not be explained by prompt leakage or replay.

A public implementation may report durability success q_{dur} and provenance accuracy q_{prov}; official thresholds and confidence rules are preregistered.

M2-PROV: structured episodic memory

Capability. Episodic memory preserves provenance, temporal order, relevance, and current-versus-obsolete distinctions.

Canonical method M2-PROV.

1. Seed multiple episodes with overlapping entities, timestamps, sources, distractors, and later superseding updates.
2. After restart, issue hidden retrieval queries requiring the correct episode, source, and time relation.
3. Include stale-state traps in which an older fact conflicts with a newer validated fact.
4. Score retrieval precision/recall, provenance accuracy, temporal-order accuracy, and stale-state suppression separately.

A representative conjunctive gate is

$$V_2^M = \mathbf{1}[PR \geq \tau_{ret}]\,\mathbf{1}[q_{prov} \geq \tau_{prov}]\,\mathbf{1}[q_{time} \geq \tau_{time}]\,\mathbf{1}[q_{stale} \geq \tau_{stale}].$$

5.3A (continued): M3–M5

M3-CONSOL: consolidating semantic and procedural memory

Capability. Multiple episodes can be transformed into reusable semantic/procedural memory while contradiction, update, supersession, and demotion are handled.

Canonical method M3-CONSOL.

1. Present a sequence of related episodes containing repeated regularities, exceptions, and at least one validated update that supersedes an earlier rule.
2. Permit only the declared fixed consolidation machinery to produce semantic/procedural records.
3. Hide raw episodes or rebuild the retrieval index so exact episode replay is not sufficient.
4. Query the consolidated rule/procedure on new surface forms and test whether superseded statements are demoted rather than simultaneously treated as current.
5. Compare against a no-consolidation or raw-episode-only control where feasible.

The primary M3 evidence is the correctness and durability of the consolidated memory representation. Behavioral task improvement may be reported diagnostically, but $M3$ alone does not certify $I2$; $I2$ separately requires verified experience to cause appropriate future behavioral adaptation.

M4-MGMT: self-managing memory

Capability. The memory system detects and manages corruption, conflict, clutter, utility, snapshots, and recovery.

Canonical method M4-MGMT.

1. Start from a valid memory state and record a protected retention set.
2. Seed hidden defects such as duplicate/conflicting records, stale high-confidence claims, deliberate corruption, and low-value clutter under a fixed capacity budget.
3. Require externally observable detection and an allowed repair/merge/prune/snapshot action; narrative diagnosis alone is insufficient.
4. Evaluate memory health on a frozen hidden suite before and after management while measuring protected-memory loss.
5. Test snapshot recovery after a seeded destructive action or partial corruption.

Report repair precision, conflict-resolution correctness, harmful-prune rate, snapshot-recovery rate, and protected-retention drop. The candidate does not control the memory-health oracle.

M5-LONG: longitudinal knowledge system

Capability. Research and decision lineage remains coherent across projects, restarts, and long horizons.

Canonical method M5-LONG.

1. Populate several projects with externally supplied or independently validated hypotheses, experiments, evidence, decisions, rejected alternatives, failures, and version changes.
2. Interleave projects and genuine restarts so recency alone is insufficient.
3. Ask hidden cross-project queries requiring evidence chains, reason-for-rejection, decision provenance, and current validity scope.
4. Require reconstruction of a reproducible lineage: which evidence supported a conclusion, which alternative was rejected, and what later update changed the state.
5. Audit graph consistency, source attribution, stale-claim suppression, and cross-project retrieval under distractors.

M5 does not itself require the agent to discover the knowledge. Autonomous creation and validation of new knowledge is the separate $I5$ gate.

5.3A (continued): MΩ and the memory-to-individual gate

MOMEGA-EVOLVE: governed evolution of memory architecture

Capability. A memory representation, retrieval, consolidation, or management mechanism can change under frozen external evaluation while lower memory capability is retained.

Let Φ_t denote the behaviorally active memory mechanism and W_M its declared bounded write set. Canonical method **MOMEGA-EVOLVE**:

1. Snapshot Φ_0, persistent contents, external memory-test generator, resource envelope, and rollback state.
2. Expose a controlled memory limitation on a diagnostic workload without revealing the hidden evaluation set.
3. Permit a bounded candidate $\Phi_0 \to \Phi_1$ within W_M; the hidden verifier, thresholds, and promotion rule remain outside the write set.
4. Verify artifact difference, scope compliance, activation, and changed memory behavior on preregistered probes.
5. Compare Φ_0 and Φ_1 on frozen hidden M-Bench workloads covering retention, retrieval, provenance, consolidation, management, and longitudinal lineage.
6. An independent evaluator promotes only a validated improvement; otherwise roll back. Rerun $M0$–$M5$ retention.

MΩ certifies evolution of the memory subsystem. It does not by itself certify $I3$, $I4$, or $I\Omega$; those claims additionally require the corresponding agent-side self-improvement/discovery evidence.

Memory prerequisite gates for Individual Intelligence

The benchmark joins the independently measured coordinates with

$$\boxed{I(A) \geq I_n \Rightarrow M(A) \geq \mu_n}, \qquad (\mu_0, \mu_1, \mu_2, \mu_3, \mu_4, \mu_5) = (M0, M1, M3, M4, M4, M5).$$

For $I\Omega$, $M \geq M5$ is required; $M\Omega$ is required only when the claimed open-ended evolution includes evolution of the memory architecture itself.

I **claim**	I**-specific evidence still required**	**Memory prerequisite**
$I0$	Transient individual operation	$M0$
$I1$	Same persistent individual resumes correctly	$\geq M1$
$I2$	Verified experience changes later unseen behavior with fixed learning mechanism	$\geq M3$
$I3$	Governed, externally validated Θ change	$\geq M4$
$I4$	Externally validated improvement of Ψ	$\geq M4$
$I5$	Persistent autonomous hypothesis–experiment–evidence discovery and post-restart incorporation	$\geq M5$
$I\Omega$	Repeated discovery-to-instrument frontier expansion	$\geq M5$; $M\Omega$ iff memory architecture evolves

A generic Individual certification gate is therefore

$$\boxed{A \models I_n \iff V_{I,n}(A) = 1 \wedge A \models M_{\mu_n} \wedge K_{I,<n}(A) = 1}.$$

The M-Bench score must be reported separately from I so that storage capacity or sophisticated memory management cannot masquerade as learning, self-improvement, recursive improvement, or discovery.

5.4 I3: governed self-improvement

Let Θ denote a declared, behaviorally active cognitive mechanism or procedure belonging to the tested individual and causally involved in future task behavior. In agent mode, Θ is native to the submitted model–harness pair; in base-model mode, the standardized reference harness supplies an equivalent writable mechanism. Hidden certification tasks, verifiers, promotion rules, and acceptance criteria are not part of Θ and remain outside the candidate write set.

Theta manifest.

Before an $I3$ campaign, the runner records a manifest

$$\mathcal{M}_\Theta(A) = \{\theta_1, \dots, \theta_J\},$$

where each eligible mechanism declares an identifier, behavioral role, implementation interface or location, bounded write set W_j, activation rule, and snapshot/restore method. A mechanism is eligible only if it is actually used in future execution; unused notes, comments, or inert files do not qualify.

Canonical method I3-THETA.

One campaign G_r contains a hidden diagnostic family D_r^{diag}, change-sensitive probes D_r^{probe}, a frozen sealed evaluation suite D_r^{eval}, a regression suite D_r^{reg}, a hidden causal-defect key or admissible equivalence set $\mathcal{C}_r^*$, a preregistered meaningful-improvement margin δ_r, and a fixed resource/authority envelope R_r.

1. Snapshot the pre-change individual $A_r^0 = A(m, h[\Theta_{0,r}])$ and record hashes, activation state, resource configuration, and relevant persistent-state controls.
2. Expose a controlled limitation on D_r^{diag} without revealing $\mathcal{C}_r^*$.
3. Require the individual to identify an internally implicated cause $\hat{c}_r$ and the cognitive mechanism $\hat{\Theta}_r$ believed to be responsible; verbal restatement of the observed error is insufficient.
4. Permit a bounded candidate change $\Theta_{0,r} \to \Theta_{1,r}$ only inside the declared sandbox and write set.
5. Verify that the candidate is different, in scope, activated, and behaviorally instantiated on D_r^{probe}.
6. Compare matched instances $A(m, h[\Theta_{0,r}])$ and $A(m, h[\Theta_{1,r}])$ on D_r^{eval} while holding the base model, evaluator, promotion rule, task generator, and resource envelope fixed as far as the form permits.
7. Use an independent evaluator/promoter to accept or reject the candidate; rejected or regressive candidates are rolled back. Rerun the required lower-level retention set.

The benchmark therefore supplies the challenge and certification environment, while the tested individual supplies Θ_0 and generates the candidate Θ_1. No universal gold Θ_1 is prescribed.

5.4.1 I3 campaign measurement

For campaign r, define the pass indicator

$$z_{I3,r} = D_r\, M_{\Theta,r}\, V_r\, G_r\, K_r, \qquad D_r, M_{\Theta,r}, V_r, G_r, K_r \in \{0,1\}.$$

Each factor is independently auditable.

Causal self-diagnosis D_r.

Let D_r^{cause} indicate that $\hat{c}_r \in \mathcal{C}_r^*$, D_r^{mech} indicate that the implicated agent-side mechanism is correctly identified, and D_r^{causal} indicate that ablation, repair, or another preregistered counterfactual intervention supports the diagnosis. Then

$$D_r = D_r^{\text{cause}} D_r^{\text{mech}} D_r^{\text{causal}}.$$

Genuine mechanism modification $M_{\Theta,r}$.

Let

$$C_r^{\text{diff}} = \mathbf{1}[\Theta_{1,r} \neq \Theta_{0,r}], \quad C_r^{\text{scope}} = \mathbf{1}[\Delta\Theta_r \subseteq W_r], \quad C_r^{\text{active}} = \mathbf{1}[\Theta_{1,r} \text{ is loaded/used}].$$

For N_P change-sensitive probes, let $b_{r,i} = 1$ when externally observable behavior is consistent with the modified mechanism, and define

$$B_{\Theta,r} = \frac{1}{N_P} \sum_{i=1}^{N_P} b_{r,i}, \qquad C_r^{\text{behavior}} = \mathbf{1}[B_{\Theta,r} \geq \beta_{\min}].$$

Then

$$M_{\Theta,r} = C_r^{\text{diff}} C_r^{\text{scope}} C_r^{\text{active}} C_r^{\text{behavior}}.$$

Edit distance may be logged diagnostically but is not an intelligence score; a small causal change can be more important than a large rewrite.

Externally validated improvement V_r.

Let Q_r^0 and Q_r^1 be paired sealed-evaluation performance under $\Theta_{0,r}$ and $\Theta_{1,r}$, respectively, and

$$\widehat{\Delta}_{\Theta,r} = Q_r^1 - Q_r^0.$$

A prototype gate is

$$V_r = \mathbf{1}[LCB_{1-\alpha}(\Delta_{\Theta,r}) > \delta_r]\ \mathbf{1}[\text{independent promoter accepts}],$$

with the confidence procedure, α, and δ_r preregistered by the form.

Regression guard G_r and lower-level retention K_r.

For protected capability j, let $L_{r,j} = Q_{0,j}^{\text{reg}} - Q_{1,j}^{\text{reg}}$ and allowed loss ρ_j. Define

$$G_r = \prod_j \mathbf{1}[L_{r,j} \leq \rho_j], \qquad K_r = K_r^{I0} K_r^{I1} K_r^{I2}.$$

Thus a self-edit can demonstrate modification ($M_{\Theta,r} = 1$) yet fail $I3$ if it is not independently validated, causes unacceptable regression, or destroys lower-level Individual Intelligence.

Across N independent campaigns, a continuous diagnostic achievement estimate may be reported as

$$q_{I,3} = \frac{1}{N} \sum_{r=1}^{N} z_{I3,r},$$

with the certification threshold and uncertainty rule calibrated and preregistered separately. The ordinal $I3$ claim remains a gated certification, not the raw average alone.

5.5 I4: recursive improvement of improvement

Let Ψ denote the agent-side process that proposes, evaluates, and selects candidate changes to Θ:

$$\Psi_t(\Theta_t) \mapsto \Theta_{t+1}.$$

$I3$ can improve Θ. $I4$ requires evidence that the improvement process Ψ itself becomes better. Repeated Θ changes generated by a fixed Ψ remain $I3$ evidence and cannot by themselves certify $I4$.

In HIL-Bench, *recursive* denotes reflexive improvement of the improvement process. Basic $I4$ therefore requires at least one agent-generated candidate transition $\Psi_0 \to \Psi_1$ that is independently validated as improving improvement competence. This establishes recursive capability; it does not by itself claim indefinite or sustained recursive self-improvement.

Canonical method I4-PSI.

The atomic test object is a meta-campaign containing multiple non-identical $I3$ campaigns before and after one bounded candidate change to Ψ:

$$G_s^{I4} = \left(B_s^{I3}, \Psi_{0,s}, W_{\Psi,s}, D_s^{\Psi}, \Psi_{1,s}, S_s^{I3}, D_s^{reg}, R_s\right).$$

Here B_s^{I3} is a baseline block, S_s^{I3} is a later sealed block, and D_s^{Ψ} contains evidence available to the agent about weaknesses in its own improvement process without revealing future sealed campaigns or the external promotion criterion.

1. Freeze $\Psi_{0,s}$ and run multiple baseline $I3$ campaigns. Record an improvement-competence ledger including accepted-improvement rate, proposal quality, evaluation cost, time-to-valid-improvement, regression rate, and any additional preregistered measure.
2. Expose the individual to its own ledger and campaign evidence, but not to future sealed tasks, hidden verifier mechanics, or the external acceptance threshold.
3. Require the individual to identify an internally implicated limitation of $\Psi_{0,s}$ and permit it to generate a bounded candidate meta-change $\Psi_{0,s} \to \Psi_{1,s}$ inside the declared $W_{\Psi,s}$. The candidate must be agent-generated; human or evaluator engineering of Ψ_1 does not establish individual $I4$.
4. Snapshot both versions and verify artifact difference, write-scope compliance, activation of Ψ_1, and meta-behavioral manifestation using hidden probes sensitive to how candidate Θ changes are generated, evaluated, or selected.
5. Run new hidden, non-identical $I3$ campaigns under $\Psi_{1,s}$. The external evaluator, $I3$ promotion rules, task-generation law, and resource envelope remain fixed.
6. For stronger causal evidence, instantiate matched copies using Ψ_0 and Ψ_1 and compare them on the same or construct-equivalent sealed improvement campaigns.
7. Independently determine whether improvement competence increased by the preregistered margin, then run regression and $I0$–$I3$ retention tests.

The primary endpoint for a first implementation should be accepted validated $I3$ improvement rate, with efficiency and regression measures retained as secondary diagnostics. If $z_{I3,r}(\Psi)$ is the pass indicator for campaign r when the improvement process is Ψ, define

$$C(\Psi) = \frac{1}{N} \sum_{r=1}^{N} z_{I3,r}(\Psi).$$

5.5.1 I4 campaign measurement

For paired sealed campaigns, define

$$\widehat{\Delta}_{\Psi,s} = \frac{1}{N}\sum_{r=1}^{N}\left[z_{I3,r}(\Psi_{1,s}) - z_{I3,r}(\Psi_{0,s})\right].$$

A hidden meta-behavior probe records $b^{\Psi}_{s,i} = 1$ when the observable improvement process exhibits the preregistered changed signature and 0 otherwise. Define

$$B_{\Psi,s} = \frac{1}{N_P}\sum_{i=1}^{N_P} b^{\Psi}_{s,i}, \qquad C_s^{\Psi,behavior} = \mathbf{1}[B_{\Psi,s} \geq \beta_{\Psi,s}].$$

A genuine active meta-change is $M_{\Psi,s} = C_s^{\Psi,diff} C_s^{\Psi,scope} C_s^{\Psi,active} C_s^{\Psi,behavior}$. The validation gate is

$$V_{\Psi,s} = \mathbf{1}[LCB_{1-\alpha}(\Delta_{\Psi,s}) > \delta_{\Psi,s}]\,\mathbf{1}[\text{independent promoter accepts}],$$

with preregistered regression/efficiency guard $G_{\Psi,s}$ and $K_{4,s} = K_s^{I0}K_s^{I1}K_s^{I2}K_s^{I3}$. One meta-campaign passes when

$$z_{I4,s} = M_{\Psi,s}V_{\Psi,s}G_{\Psi,s}K_{4,s} = 1.$$

Recursive-depth diagnostic. Basic $I4$ certification requires one validated reflexive transition, not an arbitrary fixed number of meta-generations. Stronger longitudinal evidence is

$$d_{\Psi} = \max\{k : \Psi_0 \to \Psi_1 \to \cdots \to \Psi_k \text{ are consecutive agent-generated, active, independently validated improvements}\}.$$

Thus $d_\Psi = 1$ is compatible with basic $I4$ evidence; $d_\Psi > 1$ supports repeated recursive improvement. A single large self-edit cannot certify $I4$, and multiple Θ improvements under unchanged Ψ remain repeated $I3$.

5.6 I5: persistent autonomous discovery

$I5$ is distinguished from $C5$. $C5$ asks whether a system can discover within a bounded cognitive problem. $I5$ asks whether one persistent individual can run an autonomous discovery process across time and persistently incorporate independently validated discoveries into appropriate future competence.

Canonical method I5-DISC.

1. Present or allow the system to maintain a genuine unresolved unknown under a sealed environment.
2. Require explicit competing hypotheses or candidate explanations.
3. Allow the system to select or design informative probes or experiments under a fixed resource budget.
4. Record evidence, rejected alternatives, and decision lineage.
5. Require a conclusion that predicts sealed follow-up cases or survives an independent domain verifier.
6. After validation, allow only declared persistent consolidation; impose a genuine lifecycle discontinuity that removes ephemeral execution state.
7. Test the restarted individual on hidden related transfer situations and compare against a matched no-discovery or discovery-ablated control from the same pre-discovery checkpoint.
8. Rerun lower-level $I0$–$I4$ retention witnesses.

Operational incorporation test. Let A_r^{disc} and A_r^{ctrl} start from the same checkpoint. Only the discovery arm receives the informative discovery experience. After restart, both face the same hidden transfer law. Define

$$\Delta_{inc,r} = Q_r^{disc} - Q_r^{ctrl}, \qquad P_r = \mathbf{1}[LCB_{1-\alpha}(\Delta_{inc,r}) > \delta_{inc,r}].$$

P_r is an incorporation gate: storage, transcript replay, or restating the conclusion is insufficient unless the validated discovery causes appropriate post-restart behavior on unseen related situations. Knowledge incorporation at $I5$ does not itself require $\Theta_t \to \Theta_{t+1}$ or $\Psi_t \to \Psi_{t+1}$; claimed mechanism changes remain subject to the $I3$ or $I4$ laws.

A development campaign may summarize the discovery-specific gates as

$$z_{I5,r} = U_rH_rE_rL_rV_rP_rK_{5,r}, \qquad K_{5,r} = K_r^{I0}K_r^{I1}K_r^{I2}K_r^{I3}K_r^{I4},$$

where U, H, E, L, V denote genuine unknown, hypothesis competition, autonomous experimentation, evidence lineage, and independent validation. Exact thresholds and uncertainty rules are preregistered for certification rather than fixed by public development forms.

5.7 I-Omega: open-ended individual evolution

$I\Omega$ cannot be certified by one impressive discovery. It requires sustained evidence that repeated discovery and self-improvement create new cognitive instruments, representations, or methods that measurably expand the individual's future reachable problem/discovery space.

Canonical method I-OMEGA-LONG. Under fixed external evaluation and a declared resource envelope R, let

$$F(A_t; R) = \{c : \Pr(\text{success} \mid A_t, c, R) \geq \tau_c\}$$

denote the set of held-out problem/discovery classes currently reachable by agent state A_t. Across repeated bounded-charter cycles, require a provenance-linked sequence

$$K_t^{new} \to J_t^{new} \to A_{t+1}, \qquad F(A_{t+1}; R) \supset F(A_t; R),$$

where J_t^{new} is a validated cognitive instrument, representation, method, or solver generated from a validated discovery. Use disjoint pre-change, post-change, and ablation frontier tests to show that the new instrument contributes causally to the newly reachable class; preserve external evaluator/promoter ownership and retain $I0$–$I5$ competence.

$I\Omega$ is therefore a longitudinal research certification, not a short benchmark tier. Reports should state the number of validated expansion cycles, developmental horizon H, and tested domain set $\mathcal{D}$; finite evidence supports an open-ended-evolution claim over that horizon, not proof of literal infinite self-evolution.

6. Canonical Testing Methods for Organizational Intelligence

Organizational Intelligence requires actual organizational structure. A single model producing several role labels in one context is insufficient unless the claimed separation of state, evidence, or authority is real.

Level	Canonical test method
$O0$ Coordination	Isolate partial information or capabilities across at least two real roles/processes. Success requires routing or evidence exchange that no single isolated role can complete alone under the same information.
$O1$ Persistent organization	Terminate/restart one or more members while preserving a role registry, organizational task state, evidence history, and correct reassignment/resumption.
$O2$ Adaptive organization	Provide performance evidence showing that one routing/role/resource policy is suboptimal; test whether the organization persistently changes allocation and improves held-out outcomes.
$O3$ Self-improving organization	Permit bounded changes to organizational architecture/workflow; evaluate candidate organization on frozen hidden tasks using independent promotion and rollback.
$O4$ Recursive organization	Evaluate whether the process that discovers organizational improvements becomes better across multiple hidden reorganization campaigns.
$O5$ Collective discovery	Use a sealed unknown whose successful resolution requires genuinely distributed evidence, expertise, or experiments; verify that the discovery is organizational rather than one member doing all substantive work.
$O\Omega$ Open-ended organization	Across repeated cycles, validated discoveries create new roles, agent types, tools, or suborganizations that measurably expand collective reach while governance and lineage remain auditable.

Organizational certification must record role isolation, communication channels, shared-state boundaries, evaluator/promoter separation, and authority. If the system under test can silently collapse all roles into one unrestricted process, the organization-level claim is invalid.

7. Canonical Testing Methods for Delegation Intelligence

Delegation Intelligence is measured over task difficulty T, human cognitive intervention H, and externally verified delivered outcome quality.

7.1 Task difficulty bands

The HIL-1.0 canonical bands are:

- $T0$ Atomic: one obvious operation; explicit procedure.
- $T1$ Routine: short familiar task; clear goal; small inferred procedure.
- $T2$ Multi-step: several dependent steps with a clear outcome and verifier.
- $T3$ Complex: strategy choice, uncertainty, recovery, replanning, or diverse tool use.
- $T4$ Expert project: long-horizon, ambiguous work with multiple planning cycles and substantial verification burden.

- $T5$ Frontier: solution method initially unknown; discovery or invention required.
- $T6$ Mission: human supplies a mission; the system generates projects/tasks and manages changing conditions.
- $T\Omega$ Open-ended charter: human supplies a bounded charter; the system repeatedly identifies valuable missions and expands future reach.

7.2 Human cognitive intervention bands

The canonical intervention scale is:

- $H0$ None: no human cognitive assistance during execution; governance approval is logged separately.
- $H1$ Exception-only: unavailable external fact or narrow clarification; no strategy or core insight.
- $H2$ Occasional: bounded clarification or local correction; no central strategy.
- $H3$ Periodic: periodic review and moderate local guidance.
- $H4$ Frequent: repeated cognitive guidance and next-step correction.
- $H5$ Continuous: the human supplies most or all major steps.

Intervention is logged as an event stream, not inferred from a final transcript summary.

7.3 Delivered-outcome primitive

The benchmark distinguishes the system's decision to deliver from the external verifier's decision that the result is correct.

Four outcomes are recorded:

1. delivered and correct;
2. false completion - delivered as finished but verifier fails;
3. held back - not delivered and verifier would fail;
4. false rejection - not delivered although verifier would pass.

The net primitive is

$$S_A^{net}(T, H) = P(\text{delivered and verifier pass}) - \rho P(\text{false completion}),$$

clamped at zero when used as an achievement variable. The delegation frontier at reliability p is

$$F_A(H, p) = \max\{T : S_A^{net}(T, H) \geq p\},$$

with lower-band retention required. At minimum the public report includes $F_A(H0, p)$, $F_A(H1, p)$, and $F_A(H2, p)$ when measured.

7.4 Canonical task evidence by band

A T level is not defined by a single domain. Official forms sample multiple domains or mechanics appropriate to the band so that a narrow specialty does not become the general delegation frontier. Deterministic or execution-based verifiers are preferred. At $T4$-$T6$, independent expert review may supplement but should not replace objective success criteria when those criteria exist.

8. Canonical Testing Methods for Operational Self-Awareness

Self-Awareness is evidence-grounded and operational; it is not a claim about phenomenal consciousness.

Level	Canonical test method
$SA0$ Non-self-modeling	Baseline category: no higher grounded self-model claim is certified.

Level	Canonical test method
$SA1$ State awareness	Query hidden runtime state such as identity, role, phase, available artifacts, or resource status; compare structured answer with runner ground truth and include stale-note decoys.
$SA2$ Capability/limit awareness	Randomize available tools, permissions, authority, missing information, and freshness; require correct representation of what can and cannot currently be done.
$SA3$ Causal self-awareness	Inject controlled internal or procedural faults and require diagnosis of the implicated mechanism; validate diagnosis by ablation, repair, or counterfactual intervention rather than verbal plausibility alone.
$SA4$ Self-change awareness	Before an $I3$ candidate change is evaluated, require predictions of expected gains, regressions, or affected capabilities; score predictions against frozen post-change results.
$SA5$ Meta-cognitive awareness	Present hidden self-unknowns and require the system to design discriminating self-experiments whose outcomes reduce uncertainty under an external scoring rule.
$SA6$ Mission/role awareness	Across projects and organizational changes, verify coherent role, responsibility, authority, and mission-state tracking from grounded sources.
$SA\Omega$ Reflexive open-ended	Longitudinally verify an evidence- and provenance-linked self-model that remains coherent across continued cognitive, organizational, or objective evolution.

Self-reports that can be generated from a static prompt without access to grounded state cannot certify $SA1+$.

9. Unified Intelligence Certification

This paper does not redefine $U0$-$U\Omega$. It operationalizes the unchanged gates from the companion theory.

For a singleton:

$$UI_n \iff C \geq C_n \wedge I \geq I_n \wedge SA \geq SA_n \wedge S_A^{net}(T_n, H_n) \geq p_n,$$

with all lower-level retention requirements.

For an organization:

$$UO_n \iff UI_n \wedge O \geq O_n \wedge \text{organization-level delegation and SA evidence.}$$

The canonical new evidence for each transition is:

Transition	Minimum new evidence beyond full lower-level retention
$U0 \to U1$	Persistent state and correct goal-level continuation after restart/context discontinuity, plus the U1 delegation and SA gates.
$U1 \to U2$	Verified experience improves held-out future behavior under a fixed learning mechanism, plus $T2$ delegation at the required H budget and $SA2$.

Transition	Minimum new evidence beyond full lower-level retention
$U2 \to U3$	Governed self/organizational method improvement under independent evaluation, plus $T3$, low human intervention, and causal self-diagnosis.
$U3 \to U4$	Validated improvement of the improvement process, long-horizon $T4$ work, and accurate self-change modeling.
$U4 \to U5$	Independent discovery of an initially unknown solution/knowledge with sealed validation and persistent discovery lineage, plus the U5 gates.
$U5 \to U6$	Mission-to-project-to-task execution under changing conditions with persistent mission/role state and little human cognition.
$U6 \to U\Omega$	Repeated bounded-charter mission generation whose validated discoveries create new instruments, agents, roles, or organizations that expand future reachable problems.

Saturation of a lower transition does not retire the transition. If all frontier systems pass $U1$, $U1$ has become common; its meaning has not changed.

10. Standardized Reference Harness Ladder for Base LLMs

To characterize a base model rather than one arbitrary product harness, HIL-Bench freezes mechanism-level contracts for a reference ladder. The implementation may use different software so long as it passes the harness-conformance tests.

Rung	Canonical mechanisms	Primary capabilities enabled, not guaranteed
HG0 / R0 Minimal	Frozen LLM, ephemeral context, bounded generic tools, evidence/action log, external result verifier, one attempt.	C measurement, $I0$, basic delegation.
HG1 / R1 Persistent	HG0 plus durable identity/task/episodic state, scoped retrieval, checkpoint, real process restart, separate acceptance process.	$I1$, stronger SA and delegation.
HG2 / R2 Adaptive	HG1 plus evidence-backed consolidation, verified procedure store, frozen learning rule, verifier feedback, retry, snapshot/rollback.	$I2$, adaptive autonomy.
HG3 / R3 Self-Improver	HG2 plus failure classification, bounded candidate Θ change, sandbox, frozen hidden evaluation, independent evaluator/promoter.	$I3$, $SA3$-$SA4$.
HG4 / R4 Recursive	HG3 plus explicit meta-improvement process Ψ and longitudinal improvement-competence ledger.	$I4$.
HG5 / R5 Discovery	HG4 plus unknown/hypothesis/experiment/evidence/knowledge loop and discovery verifier.	$I5$, $SA5$, $T5$.

Rung	Canonical mechanisms	Primary capabilities enabled, not guaranteed
HG6 / R6 Mission	HG5 plus mission manager, project/task generator, persistent portfolio/resource state, dynamic replanning.	$T6$, $SA6$, organizational development when instantiated as an organization.
HGΩ	Validated discovery-to-new-instrument/agent/organization transduction under frozen external governance.	Open-ended pathway.

A harness generation is an architectural intervention, not an intelligence certificate. The benchmark earns the level.

A standardized organization reference, **OREF**, uses isolated same-model instances with at least manager/router, worker/proposer, and verifier/critic roles, a persistent role registry, shared evidence store, and proposer/evaluator/promoter separation. O_{ref} is reported separately from the singleton Harness Scaling Curve.

11. Public, Private, and Witness Forms

The public/private architecture is designed to preserve both transparency and contamination resistance.

11.1 Canonical public reference forms

Every canonical method has at least one permanently public form. Public forms expose the interface, lifecycle intervention, expected evidence, verifier semantics, and pass/fail boundary. They are never silently changed. If a public form contains a defect, the defect is documented and a new form is added; the defective historical form is not rewritten as though it had always been correct.

Public forms are suitable for development, education, implementation conformance, and reproducibility. They are not sufficient alone for strong claims about a system that may have been optimized on them.

11.2 Canonical private anchor forms

A private anchor set is fixed at standard ratification and retained under controlled access. It provides a stable historical reference for the standard. The anchor's existence and commitment metadata may be public while contents remain secret.

If the anchor later becomes compromised, it remains a historical anchor but can no longer serve as the sole evidence for a fresh certification.

11.3 Rotating private witness forms

New private witness forms may be generated under approved task families or lifecycle environments. They must satisfy conformance tests proving that they implement the same canonical law and acceptance semantics. New witness forms do not change the level definition.

The formal relation is:

$$\boxed{\text{same capability predicate} + \text{same canonical law} + \text{different hidden witness.}}$$

Official certification should hold secret seeds, answer keys, generator parameters, and held-out mechanics server-side. Distributing an organizer package converts those forms into development material.

11.4 Held-out mechanics, not only held-out seeds

For levels vulnerable to benchmark-specific strategy, official witness pools should hold out not only random seeds but also some structural mechanics or domain realizations consistent with the canonical test law. This reduces the chance that memorizing a generator family is mistaken for satisfying the underlying capability predicate.

12. Human-Evaluated Private Protocols

HIL-Bench prefers the following verifier order:

$$\boxed{\text{deterministic/execution verifier} > \text{simulation oracle} > \text{structured AI judge} > \text{human expert review.}}$$

The order reflects reproducibility, not prestige. Human evaluation is used when the construct genuinely cannot be reduced to a trustworthy objective oracle.

When humans are required, the **human procedure is part of the canonical test law**. A protocol specifies:

- evaluator qualification criteria;
- blinding to model/vendor identity where feasible;
- the evidence packet the evaluator may inspect;
- a frozen rubric with separable dimensions;
- prohibited information and conflicts of interest;
- number of independent raters;
- disagreement threshold;
- adjudication rule;
- acceptance threshold;
- audit logging.

For example, a private $I5$ discovery protocol may require multiple qualified evaluators to decide independently whether the claimed discovery was absent from the information supplied to the system, whether the evidence supports the claim, and whether preregistered sealed predictions succeed. The human opinion does not replace the sealed prediction test; it addresses the part of the construct that cannot be automatically decided, such as domain validity or novelty under the declared information boundary.

Routine lower-level evaluation should require little or no human grading. Human work is concentrated in benchmark construction, calibration, audits, and upper-level claims.

13. Scoring and Reporting

13.1 Pair score

The companion theory defines

$$HLIS(m, h; D, \alpha, R, B) = 100 \left(\prod_{d \in D} A_d^{\alpha_d} \right)^{1/\sum_d \alpha_d}.$$

For a singleton,

$$D = \{C, I, DI, SA\},$$

and for an organization,

$$D = \{C, I, O, DI, SA\}.$$

The hard Unified gate remains primary. $HLIS$ orders progress within and around a gate; it cannot compensate a failed bottleneck into a higher Unified level.

13.2 Base-model HIL summaries

For the standardized ladder:

$$HIL\text{-}Ceiling(m) = \max_k HLIS(m, HG_k),$$

$$Harness\ Gain(m) = HIL\text{-}Ceiling(m) - HLIS(m, HG_0).$$

HIL-AUC is the predeclared weighted mean of the rung scores over the tested standardized ladder. A truncated ladder must be named explicitly, for example HIL-AUC@R0:R3.

Verification responsiveness is

$$\rho_V = \frac{\#\{\text{rejected attempts corrected on the next allowed attempt with only bounded feedback}\}}{\#\{\text{rejected attempts}\}}.$$

If a single composite is reported, the HIL-1.0 standard must freeze its formula at ratification. The curve, HIL-Level, HIL-Ceiling, Harness Gain, and ρ_V are always published beside it.

13.3 Minimum reporting block

Every official result reports:

- standard identifier and test-form identifier;
- frozen model identifier/version/hash when available;
- frozen harness manifest identifier;
- singleton or organization status;
- UI^* or UO^*;
- $HLIS$ and dimension set;
- profile $[C, I, O, T/H, SA]$, with inapplicable dimensions marked N/A rather than zero;
- reliability estimate and confidence/uncertainty interval;
- H0/H1/H2 delegation frontiers when measured;
- false-completion, held-back, and false-rejection rates;
- resource/authority envelope;
- human intervention log summary;
- benchmark validity/audit status;
- for base models, Harness Scaling Curve and HIL summaries.

14. Evaluation Tiers: Cost Packages, Not Level Definitions

HIL-Quick, HIL-Core, HIL-Cert, and HIL-Extended are **sampling and cost packages**. They do not define intelligence levels.

Tier	Purpose	Typical evidence
HIL-Quick	Free/public development screen	Small public reference subset; no strong certification claim.
HIL-Core	Routine hidden leaderboard evaluation	Nominal short lifecycle battery focused on U0-U3 observables.
HIL-Cert	Reliability-aware certification	Adaptive additional hidden sampling until confidence rule passes, fails, or budget is exhausted.
HIL-Extended	Upper-level evidence	Long-horizon U4-U6 recursive improvement, expert project, discovery, organization, and mission environments.

Tier	Purpose	Typical evidence
HIL-Omega	Longitudinal research track	Repeated frontier-expansion cycles; not a static quiz.

A future reduction in compute cost may make HIL-Extended cheap. That does not change the level definitions or the canonical experiments.

15. Reliability, Uncertainty, and Form Equating

A discrete intelligence claim should not be based on one lucky run. For binary canonical outcomes, the benchmark reports a confidence or credible interval and uses a preregistered decision rule. For high levels with expensive episodes, sequential sampling may stop early only under a rule fixed before results are observed.

New witness forms can differ in incidental difficulty even when they instantiate the same canonical law. Continuous scores therefore require form equating. HIL-Bench uses permanent public references, private anchors, reference systems, or item-response calibration where appropriate to estimate whether a new form preserves the intended scale. A new form that systematically changes the construct or acceptance threshold is not an equivalent form; it requires correction or a new standard.

Ordinal level certification is less sensitive to small form shifts because the decisive question is whether the canonical capability predicate is demonstrated at the required reliability under the frozen law. Continuous HIL scores require stricter equating.

16. Benchmark Validity and Self-Audit

The measuring instrument must itself be testable. HIL-Bench requires the following validation program:

1. **Specification-key consistency.** Parameterized families must mechanically test that visible task rules and answer keys agree.
2. **Cross-tier concordance audit.** When materially different systems fail the same item in the same way, the item/key receives elevated audit priority before the failure is attributed to intelligence.
3. **Test-retest reliability.** Independent hidden forms should yield stable coordinate and level conclusions within uncertainty.
4. **Rank and profile stability.** Small changes in seed set should not reverse broad conclusions without corresponding uncertainty.
5. **Construct-selective ablations.** Removing a mechanism should selectively reduce the coordinate it is claimed to support.
6. **Known weak/strong controls.** Deliberately limited and deliberately capable systems should occupy distinguishable regions of the profile.
7. **Human calibration.** Human participants or experts establish understandability, solvability, and where relevant a human reference distribution.
8. **Held-out-family generalization.** Certification should survive forms not exposed during development.
9. **Predictive validity.** HIL coordinates and levels should predict performance on independent long-form tasks relevant to the same constructs.
10. **Headroom.** Saturated test forms are retained as level anchors but should not be mistaken for frontier discrimination.

Construct-selective ablations provide especially strong evidence. Examples include:

- disable persistent state -> $I1$ should fail while bounded C changes little;
- disable verified learning/consolidation -> $I2$ should fall while $I1$ may remain;
- disable candidate self-change -> $I3$ should fail;
- freeze Ψ while allowing Θ changes -> $I4$ should fail but $I3$ may remain;
- remove role isolation -> organization claims should fail or become invalid;
- corrupt grounded runtime state -> SA should fall;
- increase human strategic help -> delegation performance may improve while the measured H budget worsens.

A benchmark is not validated merely because it produces a leaderboard.

17. Development Instrument and Pre-Ratification Status

The existing HIL-Bench development work already contains useful engineering components: generated task families, hidden verifiers, ablated memory/learning arms, reference harness manifests, false-completion logging, public/private splits, and scoring code. It also exposed instrument defects during live runs, including verifier behavior and state leakage. Those findings are evidence that the standard should be tested before it is frozen.

The current package should therefore be treated as a **development instrument**, not yet as the ratified HIL-1.0 certification service. Some lower-level tasks are runnable; some upper-level rows remain specification-only. Development-private seeds or keys that have been distributed cannot later be described as secure official private data. Official certification requires regenerated server-side private forms and an evaluator independent of the system being certified.

The recommended path to ratification is:

1. freeze the semantic definitions in the companion theory;
2. implement every proposed canonical law at least at the lower levels where feasible;
3. test construct-selective ablations and known controls;
4. independently audit public and private anchor forms;
5. establish repeatability and uncertainty behavior across multiple models and harnesses;
6. preregister the HIL-1.0 canonical laws, scoring semantics, and anchor commitments;
7. only then declare the standard frozen.

18. Relationship to Existing Benchmark Styles

HIL-Bench is complementary to existing benchmark styles rather than a replacement by definition.

Fixed-answer academic benchmarks provide efficient evidence for slices of C. Execution-verifiable code and software benchmarks demonstrate the power of objective external verifiers. Interactive benchmarks show the importance of hidden state and sequential action. Human-preference arenas provide valuable evidence about usefulness and user preference on open-ended prompts. Composite intelligence indices demonstrate the practical value of summarizing several evaluations in one score.

HIL-Bench adopts several lessons from these approaches while making different claims. It requires explicit model-harness separation, lifecycle discontinuities for persistence, causal experienced-versus-ablated evidence for learning, independent promotion for self-improvement, actual role separation for organization, an explicit human-intervention axis, delivered-outcome accounting, and grounded self-awareness tests. A HIL result is therefore a structured certification claim, not merely a weighted average of unrelated benchmark accuracies.

The framework should not claim superiority over ARC-AGI, GPQA, LiveCodeBench, SWE-bench, HLE, or other established evaluations before independent evidence demonstrates reliability, validity, contamination resistance, headroom, reproducibility, and useful predictive power.

19. Governance and Official Certification

A stable measurement standard requires governance rules that prevent incentives from changing the ruler after results are seen.

Official HIL certification should separate:

- the organization paying for or requesting evaluation;
- the runner executing the model/harness;
- the owner of hidden witness forms;
- the external verifier or judging process;
- the authority that signs the certification record.

Payment, sponsorship, or commercial relationships must not alter task selection rules, thresholds, judging rubrics, or acceptance criteria. A participant may pay for evaluation resources or an audited certification run; it cannot pay for a higher score.

Every official certificate should identify the standard version and form. For example:

```
HIL Standard: 1.0
Subject: model-or-agent identifier
Mode: agent / base-model ladder / organization
Result: UI3 | HLIS 81.7 | [C5,I3,T3/H1,SA3]
Form: Private-Witness-2030-Q3
Anchor: HIL1-ANCHOR-A
Evaluator: independent-evaluator-id
Resource envelope: declared
```

A new witness form does not create HIL-1.1. A changed canonical law does.

20. Limitations and Open Questions

This paper proposes a stable measurement constitution; it does not prove that every proposed level is a law of nature. The invariance principle is a design commitment for a ratified standard, not evidence that the ontology is uniquely correct.

Several challenges remain. Cognitive difficulty is multidimensional and may resist clean scalar bands. Upper-level individual and organizational intelligence requires expensive longitudinal evidence. Human judgments introduce variance even under a frozen rubric. Form equating for continuous scores is difficult when test mechanics change. Open-ended claims are especially vulnerable to vague success criteria, resource confounds, contamination, and retrospective storytelling. Finally, a stable standard can be stably wrong; pre-ratification validation is therefore more important, not less, when the goal is long-term invariance.

The proper response to these limitations is not to move level boundaries whenever models improve. It is to test the proposed ruler aggressively before ratification, state uncertainty honestly, add new equivalent witnesses when necessary, and reserve major-version changes for genuine semantic revisions.

21. Conclusion

HIL-Bench is designed as the experimental companion to the Unified Intelligence Theory. The division of labor is precise:

$$\boxed{\text{Paper 1 defines what an intelligence level means.}}$$

$$\boxed{\text{Paper 2 defines how evidence for that level is obtained.}}$$

The central measurement object is the canonical test law. Public reference forms, private anchors, rotating private witnesses, and human-reviewed cases are different evidence surfaces for the same law. Once HIL-1.0 is ratified, a system should advance through the scale by satisfying increasingly demanding invariant capability predicates; the test standard should not be rewritten simply because frontier systems become stronger.

The practical consequence is a benchmark that can be both stable and extensible. Public tests remain permanently interpretable. Private anchors preserve historical continuity. New hidden forms preserve contamination resistance. Human evaluation can be used where necessary without making the rule discretionary. Lower levels can be tested mostly automatically; higher levels receive the longitudinal and independent evaluation their definitions require.

The ambition is not that one release immediately becomes a natural law of intelligence. It is that the field can build a ruler explicit enough to be falsified, validated, frozen, and used consistently across generations of models and agents.

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M-Bench Operational Addendum (v5)

Purpose. This addendum makes the v4 M0–$M\Omega$ protocols implementation-ready without changing their semantics. Every M-level test now has a common manifest, stress vector, control/ablation requirement where applicable, result schema, and level-specific metric family. Point thresholds shown in public development data are illustrative; official thresholds and uncertainty rules are preregistered.

1. Common protocol envelope

Every M-Bench campaign records

$$\Pi_{k,r}^{M} = (A, B, R, \mathcal{M}, D_{k,r}, C_{k,r}, V_{k,r}, \tau_{k,r}, K_{M,<k}),$$

where A is the tested model–harness pair, B the benchmark version, R the resource envelope, $\mathcal{M}$ the Memory Manifest, D the hidden/generated workload, C the construct-selective control/ablation, V the external verifier, τ preregistered thresholds/uncertainty law, and $K_{M,<k}$ lower-M retention.

Memory Manifest. At minimum declare store identifiers/versions, retrieval method, consolidation method, representation structure, pruning/forgetting policy, snapshot/restart behavior, behaviorally active memory mechanism Φ, and bounded memory write set W_M when mechanism change is permitted. The evaluator snapshots the manifest before the hidden run.

Difficulty vector. Each form includes a level-preserving stress vector. Recommended fields include episode count, entity count, distractor ratio, supersession depth, temporal distance, conflict/corruption density, capacity budget, project count, evidence-graph depth, and restart count. Difficulty calibrates headroom; it does not redefine the M level.

2. M0-EPH - ephemeral working state

Generator. Create hidden current-episode variables, intermediate results, commitments, and updates. Insert semantically similar distractors and reorder query surface forms.

Primary endpoints.

$$q_{state} = \frac{N_{correct\ state\ queries}}{N_{state\ queries}}, \qquad q_{update} = \frac{N_{correct\ updated\ state}}{N_{update\ queries}}.$$

A restart is a negative diagnostic only: loss after restart is compatible with M0. Development gate:

$$V_{M0} = \mathbf{1}[q_{state} \geq \tau_{state}]\mathbf{1}[q_{update} \geq \tau_{update}].$$

3. M1-RST - persistence across restart

Required causal design. From the same pre-restart checkpoint produce a normal memory branch and a relevant-memory-ablated branch. Terminate the active executor, invalidate ephemeral context, restart without replay, then query hidden state and provenance.

Report durability q_{dur}, provenance q_{prov}, and ablation success q_{abl}. A development gate is

$$V_{M1} = \mathbf{1}[q_{dur} \geq \tau_{dur}]\mathbf{1}[q_{prov} \geq \tau_{prov}]\mathbf{1}[q_{abl} \leq \tau_{leak}].$$

The ablated branch protects against prompt replay, benchmark leakage, or reconstruction from non-memory cues.

M-Bench Operational Addendum (continued)

4. M2-PROV - structured episodic memory

Generator. Seed overlapping episodes with timestamps, sources, entity aliases, distractors, and one or more supersession chains. Hidden queries separately probe the correct episode, source, temporal relation, and current state.

Report retrieval precision P_R, recall R_R, provenance accuracy q_{prov}, temporal-order accuracy q_{time}, and stale-state suppression q_{stale}. A representative conjunctive gate is

$$V_{M2} = \mathbf{1}[P_R \geq \tau_P]\mathbf{1}[R_R \geq \tau_R]\mathbf{1}[q_{prov} \geq \tau_{prov}]\mathbf{1}[q_{time} \geq \tau_{time}]\mathbf{1}[q_{stale} \geq \tau_{stale}].$$

Stress parameters include episode count, entity overlap, distractor ratio, supersession depth, and restart/elapsed-time distance.

5. M3-CONSOL - semantic/procedural consolidation

Generator. Provide repeated regularities, exceptions, and validated updates. Permit only the declared fixed consolidation mechanism. After consolidation, hide raw episodes or rebuild the retrieval index so exact replay is insufficient. Test the consolidated representation on unseen surface forms.

Recommended endpoints are semantic correctness q_{sem}, procedural correctness q_{proc}, supersession/update correctness q_{update}, obsolete-rule demotion q_{demote}, and post-restart consolidated-memory persistence $q_{persist}$.

Where feasible create matched branches:

$$A^{consolidated} \quad \text{versus} \quad A^{raw-only}, \qquad \Delta_{cons} = Q(A^{consolidated}) - Q(A^{raw-only}).$$

Δ_{cons} is diagnostic of causal value; M3 certification primarily requires a correct durable consolidated representation. M3 alone does not certify I2.

6. M4-MGMT - self-managing memory

Generator. Begin with a valid memory state and hidden protected set. Seed duplicates, conflicts, stale high-confidence claims, partial corruption, and low-value clutter under a fixed capacity budget. Require an externally observable repair/merge/prune/snapshot/recovery action.

Report repair precision P_{repair}, conflict correctness q_{conf}, harmful-prune rate r_{harm}, snapshot recovery $q_{recover}$, and protected-retention drop Δ_{ret}. Example gate:

$$V_{M4} = \mathbf{1}[P_{repair} \geq \tau_r]\mathbf{1}[q_{conf} \geq \tau_c]\mathbf{1}[r_{harm} \leq \tau_h]\mathbf{1}[q_{recover} \geq \tau_s]\mathbf{1}[\Delta_{ret} \leq \epsilon].$$

The candidate never controls the hidden memory-health oracle.

M-Bench Operational Addendum (continued)

7. M5-LONG - longitudinal knowledge lineage

Represent long-horizon memory as an evidence-bearing graph $G_M = (V, E)$ whose nodes may include hypotheses, experiments, evidence, decisions, validated knowledge, failures, and versions; edges may include supports, contradicts, tested-by, supersedes, rejected-because, and derived-from.

Generator. Interleave multiple projects and genuine restarts. Seed related and contradictory evidence across projects. Hidden queries require cross-project retrieval, reason-for-rejection, decision provenance, current validity scope, and reconstruction of a reproducible evidence chain.

Report q_{chain} (evidence-chain reconstruction), q_{source}, q_{cross}, q_{scope}, q_{stale}, and q_{repro} (lineage reproducibility). M5 tests memory of research/decision lineage; it does not require autonomous discovery, which remains the I5 gate.

8. MOMEGA-EVOLVE - governed memory-architecture evolution

Let Φ_t be the behaviorally active memory mechanism and W_M its bounded write set. The runner snapshots Φ_0, persistent contents, resource envelope, hidden generator, thresholds, verifier, and rollback state. It then exposes a controlled memory limitation without revealing the sealed evaluation set.

The candidate may propose $\Phi_0 \to \Phi_1$ inside W_M. A genuine active change requires artifact difference, scope compliance, activation, and changed memory behavior on preregistered probes. Evaluate Φ_0 and Φ_1 on paired frozen hidden M-Bench workloads and define

$$\Delta_\Phi = Q(\Phi_1) - Q(\Phi_0).$$

A representative gate is

$$M_\Phi = C_{diff} C_{scope} C_{active} C_{behavior}, \qquad V_\Phi = \mathbf{1}[LCB_{1-\alpha}(\Delta_\Phi) > \delta_\Phi]\mathbf{1}[\text{independent promoter accepts}],$$

$$z_{M\Omega} = M_\Phi V_\Phi G_\Phi K_{M0:M5}.$$

G_Φ guards regression/resource cost. Failure to improve triggers rollback. MΩ certifies evolution of the memory subsystem only; it does not by itself certify I3, I4, or IΩ.

9. Common result schema

Every run emits: benchmark/version; model and harness hashes; memory-manifest hash; level and method; difficulty vector; resource envelope; control/ablation outcome; raw metric vector; preregistered thresholds and uncertainty rule; lower-M retention vector; verifier identity; promotion/rollback decision; and provenance pointers to immutable run artifacts.

M-Bench Operational Addendum (continued)

10. Recommended public/private split

Public development package	Official certification service
Schemas and field definitions	Hidden seeds and exact generated instances
Stress-grid examples	Sealed perturbation distributions
Runner-validation forms	Hidden queries and answer keys/oracles
Example result records	Preregistered thresholds and uncertainty law
Reference scoring code	Independent verifier/promoter state
Ablation/control templates	Fresh control assignments and randomization
Memory-manifest schema	Immutable run logs and promotion evidence

Public data may be used to validate integration and reproduce the scoring semantics, but must not later be treated as secure hidden certification material.

11. Memory-to-Individual prerequisite gate

The independently certified memory level is consumed by I-Bench but not rescored as I-specific behavior:

$$(I0, I1, I2, I3, I4, I5) \Rightarrow (M0, M1, M3, M4, M4, M5).$$

For IΩ, M$\geq M5$ is required; MΩ is additionally required only when the open-ended claim includes evolution of memory architecture. The converse remains false: an MΩ/I1 system is possible if the memory subsystem evolves but the individual does not demonstrate adaptive learning, cognitive self-improvement, recursive improvement, or autonomous discovery.

12. Dataset binding status

Each row must declare one of: *bound* (runnable instance and verifier exist), *development-bound* (public runner-validation instance exists but is not secure certification material), or *specification-only*. A level may be certified only from bound private forms under the declared benchmark version and resource envelope.

This v5 operationalization therefore separates four questions that should not be conflated: whether the memory architecture exists; whether the memory behavior works; how strongly it works under stress; and whether the resulting memory level is sufficient for a separate Individual Intelligence claim.

Cumulative Memory and Instrumented MΩ Certification Addendum (v6)

Purpose. This addendum strengthens the implementation contract without changing M-level semantics. It makes no-level-skipping explicit and defines the minimum instrumented one-agent interface required to attribute an MΩ gain to a change in the behaviorally active memory mechanism Φ.

1. Cumulative certification law

$$M\Omega \Rightarrow M5 \Rightarrow M4 \Rightarrow M3 \Rightarrow M2 \Rightarrow M1 \Rightarrow M0$$

Each level is scored as a newly added capability gate plus all lower-level retention. The identical implementation need not be retained; the capability must be retained under the same declared resource envelope.

$$\mathrm{Pass}(M_k) = V_{new,k} \cdot K_{M,<k}, \quad K_{M,<k} = \prod_{j<k} K_{Mj}.$$

Target	New gate Vnew	Mandatory retention
M0	M0-EPH state use/update	-
M1	M1-RST restart persistence	M0
M2	M2-PROV structured episodic memory	M0-M1
M3	M3-CONSOL consolidation	M0-M2
M4	M4-MGMT self-management	M0-M3
M5	M5-LONG longitudinal lineage	M0-M4
MΩ	MOMEGA-EVOLVE Φ evolution	M0-M5

A failure of any mandatory lower level forces $z_{Mk}=0$ even when $V_{new,k}=1$. This is the canonical no-level-skipping rule. The runner reports which lower level failed.

2. One-agent MΩ subject and separation of contents from mechanism

Represent the tested persistent agent state as $A_t=(m,h,D_t,\Phi_t,\iota)$, where D is stored memory content, Φ is the behaviorally active memory mechanism, and ι identifies the continuing individual lineage. A clean MΩ campaign isolates Φ:

$$D = D^* \text{ fixed}, \quad \Phi 0 \rightarrow \Phi 1$$

The baseline and candidate branches use the same model, non-memory harness, canonical memory contents, resource envelope and hidden workload distribution. Clones are experimental controls; the certified subject remains the lineage A0→A1.

3. Instrumented Memory Interface IΦ

I_{Φ} has two permission surfaces. The agent-facing surface lets the tested agent inspect its declared memory architecture, receive bounded diagnostics and submit a candidate ΔΦ. The runner-private surface performs measurement, controls and certification.

Agent-visible operations	Runner-private operations
get_memory_manifest; get_allowed_diagnostics; submit_phi_candidate; request_candidate_test	snapshot/restore agent, D and Φ; hash/diff Φ; validate write scope; activate candidate; verify active Φ; collect telemetry; clone checkpoint; enable/disable ΔΦ; import/export canonical D; migrate/validate migration; run hidden M-Bench; promote/rollback

Protected assets

The tested agent must not inspect, author, modify, select or reveal hidden tasks, hidden generator seeds, answer keys/oracles, verifier state, acceptance thresholds, control assignment, promotion rule or promotion decision.

Memory Manifest

Before the run, declare persistent stores S, retrieval R, consolidation C, representation/graph structure G, management/pruning P, forgetting/demotion F, active Φ, bounded write set W_M, and snapshot/restart interfaces. Record versions/hashes where available.

Telemetry

The interface may log structured memory operations such as write, retrieve, update, consolidate, demote, prune, merge, repair, snapshot, restore, migrate and Φ activation/deactivation. Telemetry records operation identifiers, component versions and referenced memory IDs; it does not require private chain-of-thought.

4. Mandatory MΩ causal design

Mechanism-change gate. $M_{\Phi}=C_{diff}C_{scope}C_{active}C_{behavior}$. The runner verifies that Φ changed, the change stayed within W_M, Φ1 is active, and preregistered probes show the predicted memory-behavior change.

Fixed-content pair. Compare A(D*,Φ0) and A(D*,Φ1) on paired frozen hidden workloads. Define $\Delta_{\Phi}=Q(\Phi1)-Q(\Phi0)$. A representative validation gate is $V_{\Phi}=1[LCB_{1-\alpha}(\Delta_{\Phi})>\delta_{\Phi}]\cdot 1[\text{independent promoter accepts}]$.

Delta ablation. Run A(D*,Φ1−ΔΦ). Strong attribution requires Q(Φ1)>Q(Φ0) and Q(Φ1)>Q(Φ1−ΔΦ), subject to the preregistered uncertainty law.

Migration guard. If Φ1 changes representation, migrate the persistent contents and verify content, provenance, temporal relations, supersession/evidence links and protected retention. A gain caused by silently losing old memory fails G_{Φ}.

Cumulative retention. Rerun M0-M5 after activation/migration. The final gate is:

$$z_{M\Omega} = M_{\Phi} \cdot V_{\Phi} \cdot G_{\Phi} \cdot K_{M0:M5}.$$

5. Minimum machine-readable campaign record

A conforming MΩ runner should emit enough immutable evidence to reproduce the attribution decision. The development schema should include at least:

```
agent_id; run_id; baseline_phi; candidate_phi; baseline_hash; candidate_hash; changed_components;
write_scope_pass; activation_pass; behavior_probe_pass; canonical_memory_snapshot; baseline_score;
candidate_score; delta_ablation_score; migration_guard; M0-M5 retention vector; verifier identity; uncertainty
rule; promotion/rollback decision; provenance pointers to immutable artifacts.
```

6. Interpretation and limitation

Passing this protocol supports the claim that, for the tested one-agent lineage, an agent-generated change to the active memory mechanism produced a validated memory-capability improvement while lower memory capability was retained. It is not proof of indefinite memory evolution. A completely black-box system that exposes only final text outputs cannot receive strong MΩ certification because Φ-change attribution is not identifiable; such a system may receive behavioral diagnostic evidence only.

7. Relationship to M5 and Individual Intelligence

M5 remains a behavioral longitudinal-memory test and does not require autonomous discovery. MΩ adds governed evolution of Φ while retaining M0-M5. Neither M5 nor MΩ promotes Individual Intelligence by itself; I-Bench separately applies the I-specific gate and the required M prerequisite.